ZIMBABWE

MINISTRY OF PRIMARY AND SECONDARY EDUCATION

BIOLOGY SYLLABUS

FORMS 3 – 4

2024 - 2030

Curriculum Development Unit
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HARARE

ACKNOWLEDGEMENTS

The Ministry of Primary and Secondary Education wishes to acknowledge the following for their valued contribution in the production of this syllabus:

- The Secondary Level Biology Syllabus Development Panel
- Zimbabwe School Examinations Council (ZIMSEC)
- Representatives from Teachers' Colleges and Universities
- United Nations Children's Fund (UNICEF)
- United Nations Educational, Scientific and Cultural Organisation (UNESCO)

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1.0 PREAMBLE

1.1 INTRODUCTION

The Biology Syllabus is designed for learners in Forms 3-4. The learners are expected to acquire theory and practical skills as well as develop socially, emotionally, physically, and cognitively. The syllabus aims at balancing knowledge, understanding, and skills in order to enable candidates to become effective learners and to provide a firm foundation for such careers as in Medicine, Food Technology, Biotechnology, and Environmental Science.

1.2 RATIONALE

This Heritage Based Biology syllabus encourages learners to employ biological skills in solving real life problems and also emphasizes the link between human activities and the environment. Learners acquire knowledge and skills of inquiry that help them to examine critically issues that arise in their own lives and in the public domain, to contribute to debate and to take part in making decisions about their own health and well-being and that of the society. The learners will be assessed through a continuous assessment system in the form of project-based assessments, hands on experiences and demonstrations.

1.3 SUMMARY OF CONTENT

The content covered by this syllabus includes theory and practical skills in the broad areas of Biology such as: Cytology, Anatomy, Physiology, Genetics, Ecology, Systematics, Health and Diseases.

1.4 ASSUMPTIONS

The Heritage-Based Biology syllabus for Zimbabwe has taken a deliberate consideration of several assumptions critical for socio-economic transformation. The assumptions are based on the context of Zimbabwe's heritage, educational system, societal needs and aspirations. It therefore becomes critical to consider that learners:

- are exposed to scientific experiences
- live in diverse social contexts
- use technological devices
- are conscious of their environment
- are aware of their obligation towards health and well-being

The general assumption is that a Heritage-Based Biology syllabus can effectively integrate science and technology education, fostering a deeper understanding of scientific concepts, technological innovations and their relationship with Zimbabwe's heritage
The syllabus assumes that learners have:

- acquired skills in handling apparatus
- learnt about the hazards and safety precautions during experiments
- knowledge on the basic concepts of biological sciences
- developed an awareness and interest in the conservation of the environment
- a basic understanding of health and disease issues

1.5 CROSS- CUTTING THEMES

In order to foster competency development for further studies, life and work, the following cross-cutting priorities have been taken into consideration:

- Gender equity
- Environmental management
- ICT
- Children's Rights and Responsibilities
- Disaster Risk Management
- Health and wellbeing
- Climate Change
- Entrepreneurship

2.0 PRESENTATION OF SYLLABUS

The Forms 3 - 4 Heritage-Based Biology Syllabus is presented as a stand-alone syllabus which covers the major branches of Biology. It consists of the preamble, aims, objectives, methodology, scope and sequence, competency matrix and assessment.

3.0 AIMS

The aims are to help learners to:

3.1 appreciate the importance of biology for sustainable socio-economic development of the country
3.2 develop biological practical skills, accuracy, objectivity, integrity and enquiry
3.3 develop good practices for health and safety
3.4 apply scientific method in Biology, other disciplines and in solving everyday life experiences
3.5 recognise that the study and practice of biology are inter-related and are subject to economic, technological, social, political, ethical and cultural influences
3.6 develop interest and participate in caring for the local and global environment

4.0 OBJECTIVES

Learners should be able to:

4.1 demonstrate knowledge of biological terms, laws, facts, concepts, theories and phenomena
4.2 select and use appropriate technological instruments to collect and analyse data
4.3 plan and conduct experiments in which they may:
- identify a problem
- formulate research questions and hypotheses
- predict results of investigations base upon prior data
- identify variables and describe the relationship between them
- plan procedures to control independent variables
- collect data
- select the appropriate format such as: graph, chart, diagram and use it to summarise the data obtained
- analyse data, check it for accuracy and construct reasonable conclusions
- prepare written and oral reports of investigations

4.4 draw biological diagrams in two dimensions
4.5 carry out simple scientific calculations
4.6 translate information from one form to another
4.7 make logical decisions based on the examination of evidence and arguments
4.8 communicate biological information logically and concisely
4.9 apply biological principles in solving problems and in understanding new situations
4.10 identify the practical constraints affecting biological investigations
4.11 use biological principles, methods and techniques in value addition
4.12 apply appropriate methods of recycling and waste disposal
4.13 explain the effects of technological applications on the environment and the organisms within it
4.14 interpret the relationship between living organisms and between living organisms and their environment

5.0 METHODOLOGY AND TIME ALLOCATION

5.1 METHODOLOGY

The syllabus is based upon interactive, multi-sensory, learner centred and practical approaches. Principles of individuality, team work, wholeness and stimulation must be applied to enhance the learning – teaching process. The learners should be allowed to apply their experiences, knowledge, skills and attitudes in the learning of the subject. The following are the suggested methods:

- Experimentation
- Discovery method
- Demonstrations
- Problem solving
- Discussions
- Visual tactile
- e-learning
- Group work
- Educational tours
- Resource person(s)
- Project based learning
- Case studies
- Observations
- Simulations

5.1 TIME ALLOCATION

A time allocation of eight (8) periods of 40 minutes per week is recommended. Double periods are also recommended

6.0 TOPICS

The Heritage -Based Forms 3 -4 syllabus consists of thirteen compulsory topics listed below:

6.1 Branches of Biology
6.2 Chemicals of Life
6.3 Cells and Cellular Activities
6.4 Enzymes
6.5 Plant Science
6.6 Animal Science
6.7 Microbiology and Biotechnology
6.8 Genetics
6.9 Biodiversity
6.10 Ecosystems
6.11 Heath and Disease

7.0 SCOPE AND SEQUENCE

TOPIC	FORM 3	FORM 4
7.1 Safety, Careers and Branches in Biology	• Safety in the laboratory	• Safety labels and symbols

	• Branches of biology • Careers	
7.2 Chemicals of Life	• Constituents and identification of: -Water -Carbohydrates -Proteins -Lipids -Nucleic acids	Classification, chemical structure and uses: -Carbohydrates -Proteins -Fats -Nucleic acids
7.3 Cells and Cellular Activities	• plant and animal cell structure • Cell specialization • Cellular transport	
7.4 Enzymes	• Nature and properties of enzymes • Mode of action	• Industrial application of enzymes
7.5 Plant Science	• Nutrition • Productivity • Transport	• Reproduction • Coordination and response
7.6 Animal Science	• Nutrition • Gaseous exchange • respiration • Transport • immunity • Sexual reproduction in Humans	• Productivity • Homeostasis • Coordination and response • Endocrine system • Skeletal system
7.7 Microbiology and Biotechnology	• Characteristics, Types and economic importance of microorganisms	• Recombinant gene technology
7.8 Genetics	• Chromosomes and Genes • Monohybrid inheritance • Mutations	• Variation • Selection
7.9 Biodiversity	• Classification	• Threats and Conservation measures
7.10 Ecosystems	• Ecosystems • Natural systems • Artificial systems	• Management of ecosystems
7.11 Health and Diseases	• Health • Diseases (Infectious and non-infectious)	• Drug use and abuse

8.0 COMPETENCY MATRIX

FORM 3

8.1 TOPIC 1 SAFETY, CAREERS AND BRANCHES IN BIOLOGY

KEY CONCEPT	OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED LEARNING ACTIVITIES AND NOTES	SUGGESTED RESOURCES
8.1.1 Safety in the Laboratory	• identify causes of accidents in the laboratory	• Causes of accidents in the laboratory - Fire - Fumes - Acids and strong bases - Handling of micro-organisms - Improper handling of apparatus - Electricity	• Identifying causes of accidents in the laboratory • Listing accidents that may occur during experiments • Demonstrating proper handling of apparatus • Discussing electrical hazards	• Charts showing electrical hazards • ICT tools
	• outline laboratory safety rule • perform fire drills • memorise the local emergency numbers • make a fire guard around the laboratory • demonstrate use of the First Aid Kit	• Laboratory safety rules - Fire drills - Local emergency call numbers	• Discussing laboratory safety rules • Performing mock fire drills • Demonstrating emergency call • Preparing a fire guard around the laboratory • Demonstrating use of the contents of the First Aid Kit	• Fire extinguishers • Fire blankets • Sand buckets • Fire alert bells • Fume cupboards • Goggles and masks • Gloves • First Aid Kit
8.1.2 Branches of Biology	• identify various branches of biology	• Branches - Cytology	• Discussing the various branches of biology	• Relevant reference materials

KEY CONCEPT	OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED LEARNING ACTIVITIES AND NOTES	SUGGESTED RESOURCES
		- Anatomy - Physiology - Ecology - Genetics - Biotechnology - Microbiology - Zoology - Botany		• ICT tools
8.1.3 Careers	• list biology related careers	• Careers such as: - Medicine - Research scientists - Ecologists - Conservationists - Dieticians	• Researching on careers related to various branches of biology	• ICT tools • Career guidance booklet • Resource persons

8.2 TOPIC 2 CHEMICALS OF LIFE

KEY CONCEPT	OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED LEARNING ACTIVITIES AND NOTES	SUGGESTED RESOURCES
8.2.1 Water	• state the constituent elements of water • describe the chemical structure of water • relate the properties of water to its uses in living organisms and as a habitat for living organisms	• Constituent elements • Chemical structure • Properties -transparent -solvent -high specific heat capacity -reagent	• Naming the constituents of water • Constructing the model of a water molecule • Demonstrating the unusual behavior of water when it freezes	• Liquid and frozen water models of a water molecule • ICT tools

KEY CONCEPT	OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED LEARNING ACTIVITIES AND NOTES	SUGGESTED RESOURCES
		• Uses -photosynthesis -solvent -Cooling -habitats -reagent	• Relating the properties of water to its uses in living organisms	
8.2.2Carbohydrates Proteins, lipids and Nucleic Acids	• state the constituent elements of carbohydrates, proteins, lipids and nucleic acids • perform tests to identify reducing sugars, non-reducing sugars, proteins and lipids	• Constituent elements • Sub-units; -Glucose as sub units of starch, cellulose, glycogen -Amino acids as sub units of proteins -fatty acids and glycerol as sub units of lipids -nucleotides as sub units of nucleic acids • Identification of reducing sugars, non-reducing sugars, starch, proteins and lipids	• Identifying the constituent elements of carbohydrates, proteins, lipids and nucleic acids • Identifying - Glucose as sub units of starch, cellulose, glycogen - Amino acids as sub-units of proteins - Fatty acids and glycerol as sub-units of lipids - Nucleotides as sub units of nucleic acids • Carrying out the following identification tests: - Iodine test for starch - Benedicts test for reducing sugar - Biuret test for proteins - Emulsion/spot test for lipids	• Iodine solution • Benedicts solution • Biuret solution/sodium hydroxide and copper sulphate • Hydrochloric acid • Sodium hydrogen carbonate • Ethanol • Water • Filter paper

8.3 TOPIC 3 CELLS AND CELLULAR ACTIVITIES

KEY CONCEPT	OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED LEARNING ACTIVITIES AND NOTES	SUGGESTED RESOURCES
8.3.1 Plant and Animal Cells	• Define a cell • identify parts of the cells • outline the functions of cell organelles • compare the structures of plant and animal cells	• Structural and functional units • Plant and animal cell structure - Nucleus - Cytoplasm - Cell membrane - Vacuole - Cell wall - Mitochondria - Chloroplasts	• Defining a cell • Observing cells under the microscope and bio viewers • Drawing and labeling plant and animal cells • Comparing the structures of plant and animal cells • Discussing the functions of cell organelles	• Microscope • Bio sets and bio viewers • Print media
8.3.2 Cell Specialisation	• Identify specialized cells • Relate cell structure to function	• Palisade – photosynthesis • Root hair – absorption • Neurone – transmission of impulse • Red blood cell – transport of oxygen • White blood cell – immunity • Muscle cell – contraction • Sperm –fertilization	• Relating cell structure to function • Examining specialized cells using bio viewers, microscopes and photomicrographs • Drawing labeled specialized cells	• Bio sets • Bio viewers • Photomicrographs • Print media • Microscopes • ICT tools

KEY CONCEPT	OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED LEARNING ACTIVITIES AND NOTES	SUGGESTED RESOURCES
		• Ovum – fertilization and food reserve		
8.3.3 Cellular Transport	• define diffusion • investigate factors that affect the rate of diffusion • define osmosis • describe active transport	• Diffusion • Factors affecting diffusion rate: - Surface area/volume ratio, temperature and particle size • Osmosis - Turgor - Plasmolysis • Active transport	• Demonstrating diffusion as movement of particles in gases, liquids and solids • Describing diffusion • Demonstrating osmosis • Movement against concentration gradient	• Perfume • Potassium Permanganate • Potatoes • Visking tubing • Egg shell membrane • Onion skin • Microscope

8.4 TOPIC 4 ENZYMES

KEY CONCEPT	OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED LEARNING ACTIVITIES AND NOTES	SUGGESTED RESOURCES

8.4.1 Nature and properties of enzymes	• define enzyme • explain the properties of enzymes	• Biological catalysts • Protein nature and substrate specificity	• Discussing the concept of enzyme • Carrying out experiments to investigate properties of enzymes such as: amylase from germinating rapoko seedling and catalase from fresh plant and animal tissue (N.B. No tissues from human or wild animals)	• Rapoko or any other grain • Fresh tissues from plants e.g. peas, beans and potato • Fresh tissue from domestic animals e.g. liver • ICT tools • Models • Hydrogen peroxide
8.4.2 Mode of enzyme action	• explain enzyme action • investigate the factors that affect the rate of enzyme catalyzed reactions	• Mode of action (The lock and key hypothesis) • Factors affecting enzyme activity -temperature -pH	• Using models and animations to illustrate the lock and key concept • Investigating effects of temperature and pH on enzyme activities	• Fresh tissues from plants e.g. peas, beans and potato • Fresh tissue from domestic animals e.g. liver • ICT tools • Models • Hydrogen peroxide

8.5 TOPIC 5 PLANT SCIENCE

KEY CONCEPT	OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED LEARNING ACTIVITIES AND NOTES	SUGGESTED RESOURCES
8.5.1 Nutrition	• define photosynthesis • State the word equation for photosynthesis • Investigate conditions necessary for photosynthesis • explain the importance of photosynthesis in an ecosystem • investigate factors affecting the rate of photosynthesis • relate the structure of the leaf to its function in photosynthesis • investigate gaseous exchange in plants • describe the functions of nitrogen, phosphorus and potassium on plant growth	• Photosynthesis • Word equation • Conditions necessary: chlorophyll, light, carbon dioxide • Fate of end products of photosynthesis • Importance of photosynthesis • Factors affecting the rate of photosynthesis (light intensity; carbon dioxide concentration and temperature • Adaption of the leaf to photosynthesis • Gaseous exchange in leaves • Mineral Nutrients - Nitrogen (N) – protein synthesis and chlorophyll formation - Phosphorous (P) – synthesis of ATP and nucleic acids - Potassium (K) – ion and osmotic balance • Deficiency signs	• Describing the process of photosynthesis • Carrying out controlled experiments to show conditions necessary for photosynthesis • Discussing the fate of end products • Carrying out experiments to show the effects of these factors • Observing the external structure of leaf • Observing prepared slides of leaf internal structure • Carrying out experiments to demonstrate gaseous exchange in plants • Discussing the functions of nitrogen, phosphorus and potassium on plant growth • Carrying out culture experiments	• Apparatus for starch testing • Potted plants • Iodine solution • Hand lenses • Microscopes • Bio viewers • Lime water/bicarbonate indicator • ICT tools • Culture solutions • Plants

KEY CONCEPT	OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED LEARNING ACTIVITIES AND NOTES	SUGGESTED RESOURCES
	• describe the effects of deficiencies of nitrogen, phosphorus and potassium	- N – stunted growth and chlorosis - P – Stunted root growth and purplish leaf colouring - K – yellow and brown leaf margins and premature death (poor flowering and fruiting)		

KEY CONCEPT	OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED LEARNING ACTIVITIES AND NOTES	SUGGESTED RESOURCES
8.5.2 Productivity	• define biomass • define productivity • explain factors affecting productivity • define pest • identify types of plant pests and diseases • explain how pests and diseases affect productivity • describe methods of controlling pests and diseases • outline the advantages and disadvantages of each method	• Biomass – organic content • Increase in biomass overtime • Factors affecting productivity - Light - Mineral salts - Temperature - Water availability - Pests and diseases • Pest as an organism which reduces productivity • Pests - Tissue – eating - sap-sucking pests • Diseases - Bacterial wilt - - fungal rust	• Carrying out experiments to measure biomass • Discussing productivity in terms of increase in biomass • Discussing factors affecting productivity • observing plants affected by various pests and diseases • classifying pests	• Balances • Ovens • Print media • School grounds and neighboring fields • Preserved plant pests and affected plants
		• Reduction of yields • Management, cultural, chemical and biological	• Discussing effects of plant pests and diseases on productivity	• Resource person

KEY CONCEPT	OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED LEARNING ACTIVITIES AND NOTES	SUGGESTED RESOURCES
			• Discussing methods of controlling plant pests and diseases with specific reference to maize, cotton and tobacco	
8.5.3 Transport	• describe the general structure of a plant • distinguish between monocotyledonous and dicotyledonous plants • identify parts of the internal structure of a young dicotyledonous stem and root • compare internal structure of dicotyledonous stem and root • state the functions of xylem and phloem vessels	• external plant structure • stem, leaves, flowers, seeds and roots • Plant stem and root internal structure • Stem; Epidermis, cortex and vascular tissue and pith • Root; Epidermis, cortex, vascular tissue and root hairs • Xylem – support and transport • Phloem – transport	• Touring to observe the general structure of plants • Comparing the structures of dicotyledonous and monocotyledonous plants • Observing prepared slides showing internal structures of stem and root • Making labeled drawings of observed structures • Discussing functions of xylem and phloem • Discussing the concepts and functions of transpiration • Demonstrating transpiration • Observing leaves • Carrying out experiments to investigate the role using dyes to identify vascular tissue	• Hand lenses • Maize and bean plants • Maize and bean seeds • Microscopes • Bio viewers and bio sets • Prepared slides • Dyes • Celery plants or spinach • Microscope slides • Bio viewers

KEY CONCEPT	OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED LEARNING ACTIVITIES AND NOTES	SUGGESTED RESOURCES
	• define transpiration • describe the functions of transpiration • investigate factors affecting rate of transpiration • describe adaptions of leaves to minimize water loss • describe how wilting occurs	• Transpiration • Functions of transpiration - Water and ion movement in xylem - Cooling effect • Factors affecting rate of transpiration -surface area -stomata distribution -temperature -wind -humidity -light intensity • Adaptations -Reduction of surface area -Thickness of cuticle -Distribution of stomata -Presence of hairs • Wilting and its significance	• Discussing the concept and functions of transpiration • Carrying out experiments to investigate factors affecting the rate of transpiration • Demonstrating transpiration • Observing leaves • Carrying out experiments to investigate the distribution and role of stomata in water loss • Observing wilting plants • Explaining wilting and its significance. • Discussing translocation	• Leaves • Vaseline • Transparent plastic bags • Potted plants
	• define translocation • describe the effect of ring barking	• Translocation • effects to plants and the environment	• Explaining the process of ring barking • Explaining the effects of ring barking	• Print media • Educational touring

8.6 TOPIC 6 ANIMAL SCIENCE

8.6.1 Nutrition	• define balanced diet • list components of a balanced diet • list major sources of nutrients • describe the major function of nutrients • Test food for presence of nutrients • compare energy content of carbohydrates and lipids • relate food intake to age, sex, activity and health requirements • define malnutrition • discuss causes and effects of malnutrition • identify the main regions of the human alimentary canal and associated organs • state functions of parts of the alimentary canal	• components of a balanced diet • components of nutrients • Sources of nutrients • Functions of nutrients • Food tests • Energy content • Nutrient requirements • Malnutrition as under nourishment or over nourishment • Undernourishment of: -Vitamin A -Vitamin C -Vitamin D -Iron -Calcium -Iodine -Protein • Over nourishment -Fats - carbohydrates • Human alimentary canal • Ingestion, digestion, absorption, assimilation and egestion • Ruminants • Digestive system	• Discussing balanced diet • Carrying out food tests • Discussing sources and functions of nutrients NB. On lipids include synthesis of cell membranes • Carrying out case study on nutritional values of locally available foods • Carrying out food tests • Carrying out experiments to measure energy content in food • Discussing the relationship between food intake to age, sex, activity and health requirements • Discussing malnutrition. • Discussing diseases linked to malnutrition • Discussing effects of taking excess fats and carbohydrates.	• Refer to topic 2 • Carbohydrate and fat- rich foods • Resource persons • ICT tools • Resource person

			• drawing and identifying parts of the digestive system • Discussing functions of the parts of the alimentary canal • Dissecting a ruminant - Identifying the different stomach chambers - Relating each stomach chamber to function • Drawing the digestive system	

	• define mechanical digestion • describe peristalsis • explain the importance of chemical digestion • describe the function of amylase, protease, and lipase • explain the roles of hydrochloric acid, rennin and bile • describe structural adaption of small intestines to absorption • describe absorption of end products of digestion • state the function of the hepatic portal vein • state the roles of large intestine • define assimilation	• human digestive system • Ingestion, digestion, absorption, assimilation and egestion • Physical breakdown of food by teeth and muscles • Peristalsis • Molecule size and solubility • Breakdown of starch, proteins and fats • Hydrochloric acid – buffer, destruction of bacteria and hydrolysis • Rennin – clotting of milk • Bile – emulsification of fats • Features of small intestines: -length - folding -villi - thin lining -rich blood supply • Diffusion and active uptake • Transportation of products of digestion to the liver • Absorption of water, vitamin B formation and expulsion of faeces • Use of end products of digestion • Regulation of glucose and amino acids • Breakdown of alcohol and toxins • storage of fat, glycogen and Vitamin A and D • Types of teeth	• Drawing and identifying parts of the alimentary canal • Discussing mechanical digestion • Describing peristalsis • Discussing chemical digestion • Carrying out experiments with starch and amylase in visking tubing as a model gut to illustrate chemical digestion and absorption (testing for glucose) • Carrying out experiments on hydrolysis, clotting of milk and emulsification of fats • Analyzing features of the small intestine • Discussing the role of the hepatic portal vein • Describing the roles of large intestine • Discussing the fate of end products of digestion • Discussing the roles of the liver	• Printed media • Model • ICT • Printed media • Starch solution • Visking tubing • Amylase • Thermometer • Benedict solution • Hydrochloric acid, bile, starch, fats and rennin • Model of villi • ICT tools

	• outline the role of the liver in metabolism • describe types and functions of teeth • describe the structure of a tooth • identify causes of tooth decay • describe the proper care of teeth	• Internal and external structures of teeth • Causes of tooth decay - Action of bacteria on sugary food producing acids • Care of teeth - Remove plaque - Neutralizing acid - Destroying bacteria	**NB. Details of urea formation not required**	
		• Types of teeth • Internal and external structures of teeth • Causes of tooth decay - action of bacteria on sugary food producing acid • Care of teeth - remove plaque - neutralising acid - destroy bacteria	• Identifying incisors, canines, premolars and molars • Identifying the crown neck, root, enamel, dentine and pulp cavity • Describing causes of tooth decay • Experimenting on action of acid on tooth • Describing different ways of preventing tooth decay	• Model of a tooth • Acid • Tooth or any calcium containing material • Tooth paste • Tooth brush • Flossy • Salt • Ash/soda

8.6.2 Gaseous Exchange in Mammals	• identify parts of the respiratory system • state the functions of parts of the respiratory system • describe the mechanisms of breathing • describe the structure and role of the alveoli • describe factors which increase the efficiency of gaseous exchange • state the difference between inhaled and exhaled air • explain the effects of physical activity on breathing and blood circulation	• Nasal passages, larynx, trachea, bronchi, bronchioles, alveoli and capillaries • Inhalation and exhalation • Structure -large surface area -moist -thin -vascularised • Factors -Diffusion gradient -Thinness -Moisture -Vascularisation • Differences in temperature, volume of carbon dioxide, oxygen and water vapour • Depth and rate of breathing • Pulse rate	• Identifying parts using models • Discussing functions of parts of the respiratory system • Demonstrating breathing mechanisms using models and animations • Discussing factors which increase efficiency of gaseous exchange • Carrying out experiments to show that exhaled air contains a higher proportion of carbon dioxide • Measuring of pulse rate and breathing rates before and after exercises	• Use of models • Recommended textbooks • Printed media • ICT tools • Lime water/ Bicarbonate Indicator • Stop watch
	• describe some effects of smoking on respiratory system	• Short term effects -destruction of cilia -increased mucus production -constriction of the bronchioles • Long term effects -lung cancer -bronchitis	• Evaluating statistical data linking diseases and smoking	• ICT tools • Print media • Resource person

8.6.3 Respiration	• define aerobic respiration • state the word equation for aerobic respiration • list uses of energy in the body • define anaerobic respiration • state the word equations for anaerobic respiration in plants and animals • describe the production and effects of lactic acid on muscles during exercise	• Aerobic respiration • Word equation • Uses of energy • Anaerobic respiration in plants and animals • Word equations • Fatigue • Oxygen debt	• Discussing aerobic respiration • Carrying out experiments to show products of aerobic respiration • Discussing uses of energy in the body • Carrying out experiments to show Fermentation of sugar by yeast • Demonstrating effects of lactic acid on muscle during exercise	• Seeds • Thermos flask • Fermentation Apparatus
8.6.4 Transport in Mammals	• describe the human circulatory system	• heart, vessels, blood and valves	• describing components of the circulatory system	• Printed media • Models • ICT tools
	• state functions of the circulatory system • identify parts of the heart • describe the cardiac cycle • explain the importance of heart valves	• Transport of materials and distribution of heat and immunity • Parts of the heart -auricles/atria -ventricles -vessels -valves • Cardiac cycle • Function of heart valves • Pulmonary and systemic circulation • Arteries, veins and capillaries	• Discussing transportation of materials and distribution of heat and immunity • Describing the action of heart muscles and movement of blood • **N.B. Mentioning the terms diastole and systole not necessary** • Discussing the importance of heart valves	• Models • ICT tools • Bio sets, bio viewers • Microscopes • Prepared slides • Resource person • Slides, photomicrographs

	• relate pressure differences in the dual circulatory system to the functions of the two circuits • relate the structure of blood vessels to their function • identify possible causes of high blood pressure • identify possible causes of coronary heart disease • state components of mammalian blood and their functions • describe the movement of materials between capillaries and tissue fluid	• Genetic predisposition, stress, lack of exercise and diet • Diet, stress, smoking, obesity and high blood pressure • components -Red blood cells -White blood cells -Platelets -Plasma • Diffusion and pressure filtration	• Comparing pulmonary and systemic circulations • Examining blood vessels to compare their structures • Discussing possible causes of high blood pressure • Discussing possible causes of coronary heart disease • Analyzing statistics showing the link between lifestyle and occurrence of coronary heart diseases • Identifying red and white blood cells from prepared slides and photomicrographs • Discussing movement of materials between capillaries and tissue fluid by diffusion and pressure filtration	

8.6.5 Immunity	• explain how the body protects itself against infection • describe events leading to active immunity • Describe events leading to passive immunity • describe the effects of the human immunodeficiency virus (HIV) on the body • discuss the significance of immunization	• Skin, tears, clotting of blood, white blood cells (engulfing action and production of antibodies) • Infection leading to antibody production • Vaccination • Transfer of antibodies via placenta and breast milk, serum injection (snake bites and rabbies)	• Discussing natural defense mechanisms • Discussing natural active immunity and artificial active immunity • Discussing natural passive immunity in infants and artificial passive immunity	• Print media • Immunization card • Resource person
		• destruction of white blood cells leading to inability to resist infect • Extended programme of immunisation	• Discussing the effects of HIV on the body • Discussing the signs and symptoms of AIDS • studying the immunization card	• Immunization card • Resource persons

8.6.6 Reproduction in Humans	• describe the structure and function of the human reproductive system
	• describe gamete formation
	• compare male and female gametes
	• describe the menstrual cycle
	• outline pathway of sperm from testes to ovum
	• describe fertilization
	• describe the early development of a zygote
	• describe the role of the placenta
	• describe limitations of the placenta
	• describe the role of amniotic fluid
		• Male and female reproductive systems
		  • Sperm and ovum formation
		• Size, mobility and numbers produced
		  • Changes in uterus lining and ovulation.
		  • Fertile and infertile phases of cycle
		• Role of oestrogen and progesterone
		• Pathway of sperm
		• Fusion of male and female nuclei
		• Formation of a ball of cells which become implanted in the uterine wall
		Early development
		- Organ formation later development
		- – mass increase
		  • Passage of:
		-antibodies
		-nutrients
		- gases
		-excretory products
		• Separating foetal from maternal blood
			• Identifying parts of the reproductive system
			• Drawing and labelling the reproductive systems
			• Discussing the functions of parts of the reproductive system
			• Discussing formation of gametes
			• **N.B. Details of meiosis not required**
			• Discussing the difference sperm and ovum
			• Describing menstrual cycle
			• Outlining the pathway of sperm
			• Discussing fertilization
			• Discussing development of a zygote
			• **N.B. Details of mitosis not required**
			• Discussing the role of the placenta
				• Print media
				• ICT tools
				• Resource person
				• Print media
				• ICT tools
				• Resource person
				• Recommended textbooks
				• Resource person
				• Print media
				• Recommended textbooks

	• explain the dangers of taking drugs during pregnancy • explain causes of infertility • outline various methods of contraception • Identify causative agent of sexually transmitted infections (STI's) • describe symptoms and effects of STI's • discuss the methods by which HIV and AIDS can be controlled	• Reference to passage of toxins and certain viruses including HIV • Functions: • Amnion and amniotic fluid - Shock absorber - Preventing temperature fluctuations - lubrication • Dangers of taking drugs during pregnancy (Illicit drugs, smoking and alcohol) • causes -Hormonal - sperm quantity and quality - physical damage by STIs - cancers • Natural methods - Abstinence - Withdrawal - rhythm • Chemical: spermicides • Barriers: condom and diaphragm • Intra-uterine devices (IUDs) • Hormonal: injection and pills • Sterilization: vasectomy and tubal ligation • Causative agents : -bacteria -virus • Sign and symptoms Gonorrhea - syphilis	• Discussing limitations of the placenta • discussing the role of amniotic fluid • Discussing the dangers of taking drugs during pregnancy • Discussing causes of infertility • Discussing reliability, limitations and appropriateness of contraceptive methods • Discussing causative agents of STI's • Discussing effects on health and fertility by STIs • Discussing the spread of STI's and HIV, incurability of AIDS and the social implications • Visiting the local clinic or hospital for statistical data	

		- chancroid -HIV/AIDS -Genital warts --genital herpes • Mutual faithfulness, abstinence and the use of condoms • Contact tracing • Treatment with antibiotics for some STI's		

8.7 TOPIC 7 MICROBIOLOGY AND BIOTECHNOLOGY

KEY CONCEPT	OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED LEARNING ACTIVITIES AND NOTES	SUGGESTED RESOURCES
8.7.1 Microbiology and Biotechnology	• list the main characteristics of the microorganisms • outline the economic importance of microorganisms	• Types of micro-organisms - Viruses - Bacteria - Fungi • Role of: -microorganisms in decomposition -(Bioremediation) -yeast in the production of bread and alcohol -bacteria in yoghurt, cheese and insulin production -fungi in production of penicillin	• Observing structures of microorganisms on • Drawing and labelling the structures • Discussing the economic importance of microorganisms • Carrying out educational tours to bakeries, breweries and dairy industries and sewage works	• Print media • Microscope slides • ICT tools • Bakeries • Breweries • Dairies • Sewage works

8.8 TOPIC 8 GENETICS

KEY CONCEPT	OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED LEARNING ACTIVITIES AND NOTES	SUGGESTED RESOURCES
8.8.1 Chromosomes and Genes	• Define chromosome • define a gene • explain the major genetic terms	• Chromosome as including a long molecule of DNA • A unit of inheritance • Genetic terms -allele -dominant -recessive -genotype -phenotype -homozygous -heterozygous	• Discussing the - Structures of chromosomes and genes • Explaining genetic terms	• ICT tools • Print media
8.8.2 Monohybrid Inheritance	• describe complete dominance • predict possible outcomes of mono-hybrid crosses	• Dominant and recessive alleles • Homozygous and heterozygous • Phenotypes and genotypes • Genetic Crosses - F^1 and F^2 generations monohybrid ratio	• Discussing complete dominance • Constructing Genetic crosses and punnet squares • Demonstrating monohybrid crosses	• ICT tools • Coloured beads/beans • Print media

	• explain codominance	• Inheritance of ABO Blood group • Phenotypes groups A, B, AB and 0 • Gene alleles for blood groups I^A, I^B, I^0 • Sex chromosomes X and Y	• Constructing genetic crosses • Discussing and illustrating sex determination in humans	
8.8.3 Mutations	• Describe genetic and chromosomal mutations • discuss factors that increase rate of mutation	• Genetic -sickle cell anemia -albinism • Chromosomal -Down syndrome • Factors that increase rate of mutation - Genetic disposition - Radiation (UV, Gamma, Alpha, X – rays) - Carcinogens	• Discussing the concept and types of mutation • Discussing factors that increase the rate of mutation	• Print media • ICT tools

8.9 TOPIC 9 BIODIVERSITY

KEY CONCEPT	OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED LEARNING ACTIVITIES AND NOTES	SUGGESTED RESOURCES
8.9.1 Biodiversity	• state the five kingdoms of living organisms • state characteristic features of the five kingdoms	• Five kingdoms - Prokaryotae/Monera - Protoctista/Protista - Fungi - Plantae - Animalia • Diagnostic features	• Classifying organisms into five kingdoms • Discussing characteristic features of each kingdom	• ICT tools • Samples of organisms • Print media

8.10 TOPIC 10 ECOSYSTEMS

KEY CONCEPT	OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED LEARNING ACTIVITIES AND NOTES	SUGGESTED RESOURCES
8.10.1 Ecosystems	• Define ecosystem • list components of an ecosystem • identify soil as a key component of an ecosystem	• A self-contained system of interdependent organisms and the physical environment • Physical and biological components such as air, water, soil, light and living organisms • Role of soil	• Examining an ecosystem such as pond, forest, field, botanical gardens • Classifying components into physical and biological • Discussing the role of soil in an ecosystem	• Ponds • Fields • Forests • Botanical gardens
	• identify biological components of soil	• components such as litter, earthworms' nematodes, termites, fungi, bacteria and humus	• Extracting macro-organisms in the soil • Demonstrating that soil contains micro-organisms	• Wire gauze • Lamps • Soil samples • Alcohol • Lime water/bicarbonate indicator
	• state the role of the biological components	• Crumb structure, aeration, fertility and pH • Properties	• Discussing the role of biological components	• Soil samples

KEY CONCEPT	OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED LEARNING ACTIVITIES AND NOTES	SUGGESTED RESOURCES
	• compare properties of clay, loam and sandy soils	-Size of particles -Air content - Water holding capacity - Drainage - Leaching - Acidity/alkalinity (pH)	• Carrying out experiments to show techniques of measuring air, water, organic content, pH, water retention and drainage	

KEY CONCEPT	OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED LEARNING ACTIVITIES AND NOTES	SUGGESTED RESOURCES
8.10.2 Natural ecosystems	• describe a natural ecosystem • construct simple food chains and webs • interpret food webs • explain the loss of energy in food chains • Interpret ecological pyramids • describe nutrient cycles	• Natural ecosystems • Trophic levels: Producers, consumers and decomposers • Inter-relationships between food chains • Respiration and heat loss • Energy input and energy flow • Pyramids of numbers, biomass and energy • Carbon cycle • Nitrogen cycle	• Carrying out educational tours to a natural ecosystem • Discussing examples from local ecosystems • Constructing simple food chains and food webs • Discussing inter relationships between food chains • Discussing energy losses in food chains • Constructing and interpreting pyramids • Discussing nutrient cycles • Drawing nutrient cycles • **NB. Scientific names of bacteria not required**	• ICT tools
8.10.3 Artificial ecosystem	• describe an artificial ecosystem • compare species biodiversity in natural and	• Artificial Ecosystem • Species diversity	• Studying cultivated piece of land • Comparing species diversity	• Garden • Field

KEY CONCEPT	OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED LEARNING ACTIVITIES AND NOTES	SUGGESTED RESOURCES
	• artificial ecosystems • state problems caused by limited species diversity	• • Soil fertility • Pest problems	• • Discussing the problems	

8.11 TOPIC 11 HEALTH AND DISEASES

KEY CONCEPT	OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED LEARNING ACTIVITIES AND NOTES	SUGGESTED RESOURCES
8.11.1 Health	• define health • discuss levels of hygiene	• Health as a state of physical, mental, emotional and social wellbeing and not just the absence of disease • Personal hygiene • Domestic hygiene • Community hygiene - Waste disposal methods - Provision of clean, safe drinking water - Sanitation - Provision of health facilities such as clinics	• Discussing the concept of health • Carrying out cleaning campaigns around communities and in school • Discussing health issues	• Resource persons • Cleaning materials • ICT tools

8.11.2 Diseases	• define disease • state causes of diseases • classify diseases into infectious and non-infectious diseases • state the causative agent, mode of transmission and signs and symptoms of infectious diseases • explain ways of preventing and treating infectious diseases	• Definition of disease • causes such as: -infection, -genetic defects -chemicals -radiation -malnutrition -degenerative causes • Infectious diseases - Cholera - Malaria - Tuberculosis (TB) - Typhoid - Bilharzia • Non-infectious diseases - Deficiency diseases - Genetic diseases - Cancer • Causative agents, mode of transmission and signs and symptoms of infectious diseases **(NB. Scientific names of causative agents not required)** • Prevention and treatment of infectious diseases	• Discussing the concept of disease • Classifying diseases into infectious and non- infectious • Researching on causes, mode of transmission, symptoms and ways of treating and preventing infectious diseases • Discussing research findings	• ICT tools • Resource persons

FORM 4

8.12 TOPIC 1 SAFETY, CAREERS AND BRANCHES IN BIOLOGY

KEY CONCEPT	OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED LEARNING ACTIVITIES AND NOTES	SUGGESTED RESOURCES
8.12.1 Safety labels and symbols	• identify various safety labels and symbols • explain the meaning of various safety labels and symbols	• Labels and symbols	• Identifying various safety labels and symbols • Discussing the meaning of the various labels and symbols	• Containers with labels and symbols • Print media

8.13 TOPIC 2 CHEMICALS OF LIFE

KEY CONCEPT	OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED LEARNING ACTIVITIES AND NOTES	SUGGESTED RESOURCES
8.13.1 Carbohydrates	• State classes of carbohydrates • state the uses of glucose, sucrose, starch, glycogen and cellulose in living organisms	• carbohydrates as mono, di-, and poly-saccharides • glucose, sucrose, starch, glycogen and cellulose in living organisms	• Classifying carbohydrates • Discussing uses of mono -, di- and polysaccharides	• ICT tools • Print media
8.13.2 Proteins	• State the functions of proteins in living organisms	• Functions of proteins in living organisms	• Discussing the functions of proteins in living organisms	• Print media • ICT tools

8.13.3 Lipids	• state the functions of lipids in living organisms	• Functions of lipids in living organisms	• Discussing the functions of lipids in living organisms	• ICT tools
8.13.4 Nucleic acids	• State the functions of nucleic acids in living organisms	• Functions of DNA and RNA	• Discussing the functions of DNA and RNA	• ICT tools

8.14 TOPIC 3 ENZYMES

KEY CONCEPT	OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED LEARNING ACTIVITIES AND NOTES	SUGGESTED RESOURCES
8.14.1 industrial applications of Enzymes	• explain the use of enzymes in the production of washing powders	• Detergents/washing powder - Lipases – fat stains - Proteases – protein stains	• Discussing the uses of lipases and proteases in stain removal • Carrying out experiments with lipases and proteases to remove stains	• Washing powder • Lipases and proteases

8.15 TOPIC 4 PLANT SCIENCE

KEY CONCEPT	OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED LEARNING ACTIVITIES AND NOTES	SUGGESTED RESOURCES
7.5.1 Reproduction	• define asexual reproduction • describe natural and artificial methods of vegetative reproduction in plants • state advantages and disadvantages of vegetative reproduction • describe the structure and characteristics of wind and insect pollinated flowers • compare the different structural adaptations of insect pollinated and wind pollinated flowers • outline the process of pollination • distinguish between self-and cross pollination • describe the advantages of self and cross pollination	• Asexual reproduction in plants • Vegetative reproduction Natural methods - Rhizomes, tubers, suckers, runners, bulbs • Artificial methods - Cuttings, grafting -Resistance to disease and pests -genetic variation - survival of offspring - rate of propagation • Flower as a reproductive structure in plants • Wind pollinated and insect pollinated flowers • Pollination • Self and cross pollination • Advantages and disadvantages of self and cross pollination • Development of pollen tube	• Discussing asexual reproduction in plants • Observing vegetative structures in plants • Demonstrating various methods of vegetative reproduction • Discussing advantages and disadvantages of vegetative reproduction • Observing insect and wind pollinated flowers • Comparing insect and a wind pollinated flower • Examining pollen grains using microscopes or hand lenses • Discussing self and cross pollination • Discussing advantages of self and cross pollination	• Plant vegetative structures such as: stems, tubers, leaves, bulbs • Gardens • ICT tools • Grass • Irish potatoes • Bananas • Sugar cane • Sweet-potato stems • Onion bulbs • Strawberry plants • Flowers • Hand lenses • Microscopes • Pollen grains

KEY CONCEPT	OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED LEARNING ACTIVITIES AND NOTES	SUGGESTED RESOURCES
	• outline the process of fertilization • describe changes that occur after fertilization in a flower • describe seed dispersal • outline the importance of seed dispersal	**NB. No knowledge of double fertilization is required** • Formation of seeds and fruits • Various ways of seed dispersal • Importance of seed dispersal - Colonization of new areas - Reducing overcrowding of plants - Increasing survival chances of plants	• examining growing pollen tubes • Describing the growth of the pollen tube and its entry into the ovule followed by fertilization • Observing a variety of fruits and seeds • Relating structure of seed to method of dispersal • Discussing importance of seed dispersal	• Print media • ICT tools • Fruits • Seeds

KEY CONCEPT	OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED LEARNING ACTIVITIES AND NOTES	SUGGESTED RESOURCES
	• describe the structure of a monocot and dicot seed • describe the conditions necessary for germination	• seed structure -Testa/seed coat - embryo (radicle and plumule) - endosperm -cotyledon -hilum - micropyle Monocot seeds – maize seeds Dicot seeds – bean/pea/ground nut • Germination - Oxygen - Suitable temperature - Water Functions of enzymes - Respiration - Hydrolysis of stored food	• Identifying different parts (external and internal) of seeds • Making labelled drawings of seeds • Carrying out controlled experiments to investigate conditions necessary for germination	• Variety of fruits and seeds • Scalpel blades • Hand lenses • Bean seeds • Maize seeds • Pyrogallic acid • Drying agents such as calcium chloride • Refrigerator • Incubator
8.15.2 Coordination and Response	• define phototropism and geotropism • illustrate phototropism and geotropism in plants • Define a plant hormone	• response by plants to light and gravity • Plant hormones as a	• Carrying and experiments to illustrate phototropism and geotropism • Discussing commercial application of plant hormones	• Potted plants • Germinating seeds • Cardboard boxes • Blades • ICT tools • Petri-dishes • Cotton wool • Clinostat • Plant hormones • Fruits

KEY CONCEPT	OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED LEARNING ACTIVITIES AND NOTES	SUGGESTED RESOURCES
	• describe the application of hormones to growth and development plants	chemical substance produced in one part of a plant and transported to other parts where they initiate a response • Uses of: - Ethylene - Gibberellins - Auxin - Cytokinins		• Young plants

8.16 TOPIC 5 ANIMAL SCIENCE

KEY CONCEPT	OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED LEARNING ACTIVITIES AND NOTES	SUGGESTED RESOURCES
8.16.1 Productivity	• define food intake in relation to increase in mass • identify animal pests and related diseases • describe how animal pests and diseases affect productivity • describe ways of preventing plant pests and diseases	• Food conversion efficiency • Animal pests and diseases - Anthrax – bacteria - Foot and mouth – virus - Liver damage – fluke - Red water – tick (vector) • Reduced yields • Mechanisms -dosing -vaccination -dipping -quarantine	• Analyzing data from agriculture department • Constructing and interpreting growth curves • Observing animal pests • Discussing animal diseases • Discussing how pests and diseases affect productivity • Discussing prevention mechanisms	• Resource persons • Pest specimens

8.16.2 Homeostasis	• define homeostasis • Identify and label parts of the skin • Relate the functions of parts of the skin to temperature regulation • Describe the function of the brain in homeostasis • define excretion • list substances that must be excreted and organs involved • describe the structure of the urinary system • state the functions of the parts of the urinary system • describe the structure of the kidney • state the functions of the kidney	• The maintenance of a constant internal environment • Skin structure and functions of the parts • Temperature sensitivity, shivering, sweating, vaso-dilation and vasoconstriction • The hypothalamus • Negative feedback • Excretion as removal of toxic materials and waste products of metabolism • Kidney – urea, excess salt, excess water and toxins • Lungs – carbon dioxide and excess water • Skin – excess salts, and excess water • Structure of the urinary systems • Functions of kidney -renal artery - urethra -- ureter -bladder • Kidney structure -cortex -medulla -pelvis • Functions	• discussing regulation of water, pH, ions, glucose concentration and temperature • drawing and labelling parts of the skin • discussing the functions of parts of the skin • discussing the roles of parts of the skin I relation to temperature regulation • Discussing the role of the hypothalamus in temperature regulation • Discussing excretion • Discussing the excretory organs and the wastes they excrete • Drawing and labelling the urinary system • Discussing functions of parts of the urinary system • Discussing kidney structure and functions • **NB. No detailed structure of the nephron required** • Discussing how the dialysis machine works	• Print media • ICT tools • Resource person

		- urine formation - osmo-regulation - pH regulation • kidney failure and the use of dialysis machine		

	• Describe how the kidney dialysis machine works	• Kidney failure and the use of dialysis machine	• Discussing how the dialysis machine works	• Educational tour
8.16.3 Coordination and Response	• describe the structure of a neurone • describe a simple spinal reflex arc • outline functions of the main parts of the brain • describe the structure of the eye • describe the functions of parts of the eye • describe how the eye focuses images on the retina • Describe eye defects and their correction • Describe the structure and functions of parts of the ear	• Structure of sensory, relay and motor neurones • Receptor, sensory neurone, relay neurone, motor neurone, effector and spinal cord • Cerebrum, cerebellum, pituitary gland, hypothalamus and medulla oblongata • Eye -Front view -horizontal section • Parts of the eye • Refraction, accommodation and image formation • Short and long sightedness • Parts of the ear • Sound conduction • balance	• Drawing and labelling the structure of neurones • Demonstrating reflex actions such as knee jerk and blinking • Drawing the reflex arc • Identifying parts of the brain and discussing their functions • Drawing and labelling the front view and horizontal section of the eye • Discussing functions of parts of the eye • Demonstrating the refraction of light • Discussing eye defects and their correction • Drawing and labeling parts of the ear • Discussing functions of parts of the ear	• Print media • Bioviewer and biosets • ICT tools • Model
8.16.4 Endocrine System	• define hormone • describe the effects of adrenaline	• Hormone as a Chemical substance produced by a gland and carried by blood, which affects the activities of a target organ	• Discussing the production, transport and effect of hormones • Discussing effects of adrenaline	• ICT tools • Resource person

		• Adrenal gland, fright, fight and flight responses		
	• outline the role of insulin and glucagon • state the signs and symptoms of diabetes mellitus • explain how Diabetes mellitus can be managed	• Regulation of blood glucose level • Diabetes mellitus • Signs and symptoms: -frequent urination -tiredness - thirst - increased blood glucose level - glucose in urine • Management of Diabetes mellitus	• Discussing regulation of blood sugar • Discussing signs of diabetes mellitus • Discussing the management of Diabetes mellitus	
8.16.5 Skeletal system	• outline functions of the skeleton • Identify the main bones of the fore and hind limb of a mammal • describe types of joints • describe the types of movement permitted by joints • describe the action of muscles at a hinge joint	• Functions -Support, -protection - movement • Scapula, humerus, radius ulna, pelvis femur, tibia and fibula • Hinge joints, ball and socket joint • Hinge joint – one plane • Ball and socket joint – all planes • Antagonistic muscles (extensor and flexor muscles)	• Observing the human skeleton • Discussing functions of the skeleton • Identifying the bones on the skeleton • Observing joints • demonstrating movement permitted by joints • Demonstrating the action of extensor and flexor muscles	• Model of the skeleton • Model of antagonistic muscles

8.17. TOPIC 6 MICROBIOLOGY AND BIOTECHNOLOGY

KEY CONCEPT	OBJECTIVES **Learners should be able to:**	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED LEARNING ACTIVITIES AND NOTES	SUGGESTED RESOURCES
8.17.1 Recombinant Gene Technology	• define the term recombinant DNA • explain that genes may be transferred between cells of different species • outline the production of insulin using recombinant DNA technology • discuss potential benefits and hazards of recombinant DNA technology	• Recombinant DNA as a DNA molecule incorporating DNA from more than one species of organism • Production of insulin using recombinant DNA technology -Gene coding for human insulin recovered for bacteria -Transformed bacteria allowed to multiply producing insulin -Insulin then recovered from bacteria • Benefits and hazards of recombinant DNA technology	• Discussing formation of recombinant DNA • Discussing insulin production **NB. Only simplified outline of insulin production using recombinant DNA technology** • Debating on benefits and hazards of recombinant DNA technology	• ICT tools • Print media

8.18 TOPIC 7 GENETICS

KEY CONCEPT	OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED LEARNING ACTIVITIES AND NOTES	SUGGESTED RESOURCES
8.18.1 Variation	• identify variable features in plants and animals • draw graphs to show continuous distribution of characteristics • distinguish between continuous and discontinuous variation	• Variation - Features such as: leaf size, size of pods, height, coat colour, mass, tongue rolling, sex - Tables and graphs • Continuous and discontinuous variation	• Carrying out activities such as: -Measuring pupils' heights/masses in class -Counting number of boys and girls in class -Collecting leaves of the same plant and measuring length -to show variables in plants and animals • Constructing tables and graphs to show continuous and discontinuous variation	• Leaves • Balances • Rulers • Graph papers
8.18.2 Selection	• define evolution • outline natural selection as a possible mechanism for evolution • describe the applications of artificial selection • state the reasons for plant and animal breeding	• Evolution. • Variation, competition, selective survival and reproduction • Application of artificial selection in plants and animal breeding • Productivity, -quality of breed -resistance to drought and disease	• Discussing how evolution occurs in natural communities • Investigating application of artificial selection in plant and animal breeding in local communities and commercial farms	• Local communities • Commercial farms

8.19 TOPIC 8 BIODIVERSITY

KEY CONCEPT	OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED LEARNING ACTIVITIES AND NOTES	SUGGESTED RESOURCES
8.19.1Threats and conservation measures	• explain threats to biodiversity • describe conservation measures	• Threats to biodiversity - Deforestation - Invasive species - Habitat destruction - Climate change - Pollution • Measures -afforestation -uses of alternative sources of energy -preservation of endangered species and agents	• Discussing threats to biodiversity • Discussing conservation measures • Participating in tree planting activities	• ICT tools

8.20 TOPIC 9 ECOSYSTEMS

KEY CONCEPT	OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED LEARNING ACTIVITIES AND NOTES	SUGGESTED RESOURCES
8.20.1 Management of Ecosystem	• describe advantages of ground cover	• advantages -Top soil preservation - water retention -reduced evaporation	• Investigating effects of soil cover	• Agricultural plots
	• describe the harmful human activities on the ecosystems	• human activities: -Agricultural -mining - industrial	• Carrying out case studies of environmental problems in the local communities	• Print media • Local ecosystems

	• suggest corrective measures to harmful human activity • explain the concept of carrying capacity	- social activities • corrective measures • Limiting factors: - Water - Food - Oxygen - Space and shelter	• Discussing and implementing corrective measures • Discussing the concept of carrying capacity • Field trip	
	• describe the effects of exceeding the carrying capacity	• effects -Degradation of ecosystem; - -overgrazing - diseases - pollution	• Discussing the effects of exceeding the carrying capacity	

8.21 TOPIC 10 HEALTH AND DISEASES

KEY CONCEPT	OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED LEARNING ACTIVITIES AND NOTES	SUGGESTED RESOURCES
8.21.1 Drug use and Abuse	• define drug	• Drug as any substance administered into the body and modifies or affects chemical reactions	• Discussing the meaning of drug	• ICT tools
	• describe medical uses of drugs	• Antibiotics – penicillin • Analgesics – paracetamol – asprin • Anti-malaria – chloroquine and paludrine	• Discussing the importance of proper administration and use of medicinal drugs. • Discussing allergic reactions to drugs in some individuals	

	• describe the abuse of alcohol, tobacco, cannabis and solvents	• Physical, mental and social ill-effects - Alcohol - Tobacco - Cannabis - Solvents	• Discussing the effects of drug abuse • Carrying out campaigns against drug abuse	• Resource persons • specimens

9.0 ASSESSMENT

The Lower Secondary Biology syllabus learning area for Forms 3 – 4 shall be assessed through School Based Continuous Assessment (SBCA) and Summative Assessment (SA). These assessments shall be guided by the principles of inclusivity, practicability, authenticity, transparency, flexibility, validity and reliability. The principles are crucial for creating a supportive and effective learning environment that fosters growth and development in learners at secondary school level. Arrangements, accommodations and modifications shall be visible to enable candidates with special needs to access assessments.

This section covers the assessment objectives, the assessment model, the scheme of assessment, and the specification grid.

9.1 Assessment Objectives

By the end of the Lower Secondary Biology syllabus learning area for Forms 3 - 4, learners will be assessed on their ability to:

9.1.1 Show knowledge and understanding.
9.1.2 Handle information and solve problems.
9.1.3 Display experimental skills and investigations.

9.2 Assessment Model

Assessment of learners at Lower Secondary school level for Biology Syllabus shall be both Continuous and Summative as illustrated in Figure 1. School Based Continuous Assessment shall include recorded activities from the School Based Projects. The mark shall be included on the learner’s end of term and year report. Summative assessment at school level shall include terminal examinations which are at the end of the term and year.

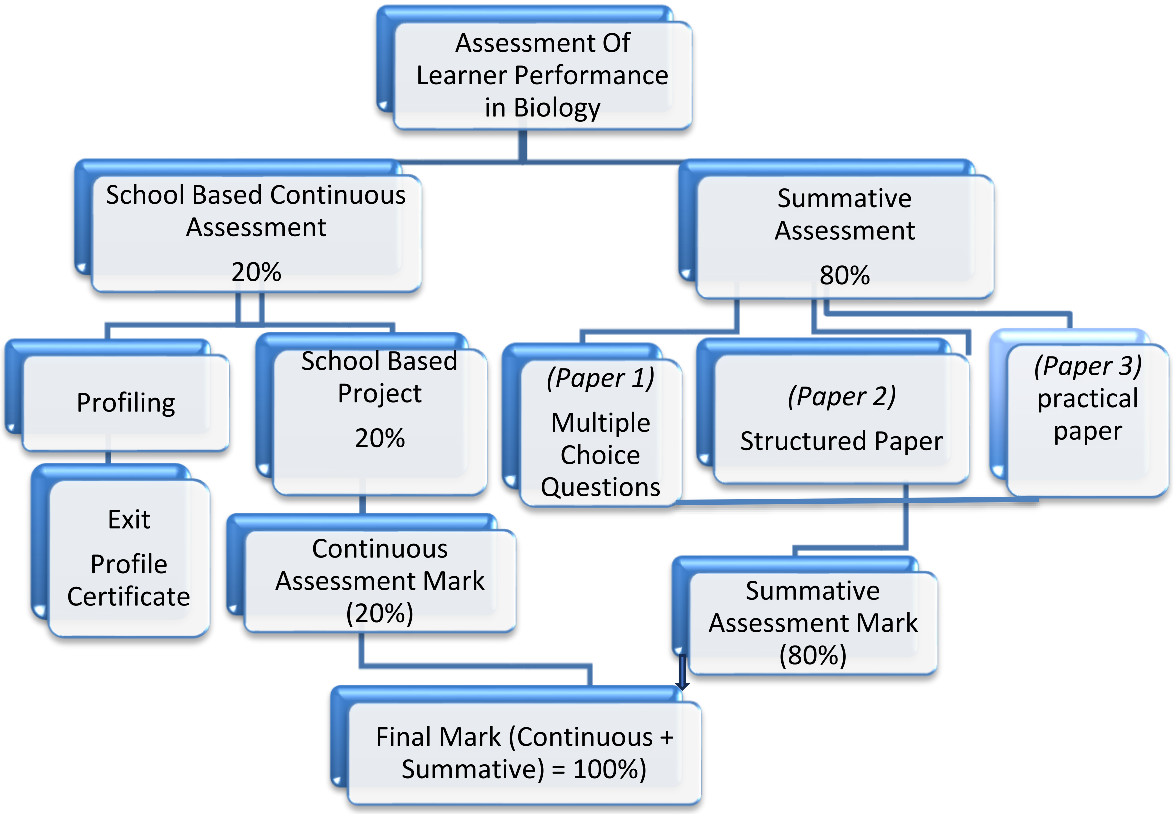

Fig. 1 Assessment Model

In addition, learners shall be profiled and learner profile records established.

Learner profile certificates shall be issued for checkpoints assessment in schools as per the dictates of the Teacher's Guide to Learning and Assessment. The aspects to be profiled shall include learner's prior knowledge, values and skills, and subsequently the new competences acquired at any given point.

9.3 Scheme of Assessment

The Assessment Model shows that learners shall be assessed using both School Based Continuous Assessment and Summative Assessment for both School and ZIMSEC assessments.

The table shows the Scheme of Assessment where 20% is allocated to School Based Continuous Assessment and 80% to School or ZIMSEC Summative Assessment.

FORM OF ASSESSMENT	WEIGHTING
School Based Continuous Assessment	**20%**
Summative Assessment	**80%**
Total	**100%**

9.3.1 Description of School Based Continuous Assessment

Learners shall do one school-based project per Form which contributes to 20% of the end of year final mark. The end of year summative assessment shall then contribute 80%. However, for ZIMSEC public examinations, two (2) school-based projects shall be considered as School Based Continuous Assessment at Form 4. The two School Based Projects shall include those done during Form 3. and Form 4 sessions. Each will contribute 10%.

9.3.1.1: School – Based Project Continuous Assessment Scheme

The Table given below shows the Learning and Assessment Scheme for the School Based Project.

Project Execution Stages	Project Stage Description	Timelines	Marks
1	Problem Identification	January	5
2	Investigation of related ideas to the problem/innovation	February	10
3	Generation of possible solutions	March	10
4	Selecting the most suitable solution	April-May	5
5	Refinement of selected solution	June	5
6	Presentation of the final solution	July	10
7	Evaluation of the solution and Recommendations	August-September	5
	TOTAL		**50**

The learning and assessment scheme shows the stages that shall be executed by pupils and the timeline at which each stage shall be carried out. Possible marks, totalling 50, are highlighted to indicate how much can be allocated.

9.3.2 Description of the ZIMSEC Summative Assessment

ZIMSEC Summative Assessment shall be a public examination at Form 4. The examination shall consist of three (3) papers of different weighting.

Paper	Paper type	Marks	Duration	Weighting
1	Multiple choice	40	1hr	20%
2	Theory	100	2hrs	40%
3	Practical test	40	1hr 30mins	20%
TOTAL				**80%**

Paper 1: Multiple choice
Duration: 1 hour
The paper consists of 40 compulsory multiple-choice items of the direct choice type. Each question shall have 4 response items.

Paper 2: Theory
Duration: 2 hours.
The paper has 2 sections
Section A will carry 40 marks and will consists of number of compulsory structured questions of variable mark valve.
Section B carries 60 marks and will consist of 4 structured questions. Each question will carry 20 marks. Candidates will be required to answer any 3 questions.

Paper 3: Practical Test
Duration: 1 hour 30mins
This paper will consist of 2 compulsory questions each carrying 20 marks.

Paper 4: School Based Project
This paper will consist of 2 projects during the 6 terms.

9.4 Specification Grid

SKILL	Paper 1	Paper 2	Paper 3
KNOWLEDGE AND UNDERSTANDING	45%	45%	25%
ANALYSIS, SYNTHESIS AND EVALUATION	40%	40%	50%
APPLICATION AND PROBLEM SOLVING	15%	15%	25%
TOTAL	**100%**	**100%**	**100%**